

Hall bounds under the Mycielski construction and the extremizers of the sixth Mycielski graph

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Abstract

For every nonempty finite simple graph G with $\chi_f(G) < 3$, we prove $\rho(\mu(G)) \leq 3$ and $\rho(\mu^2(G)) \leq 10/3$. The proof uses fractional coloring, three distinguished vertices, and elementary independence bounds; no computation enters the upper inequalities. Equality in the second bound forces an induced subset of order twenty and independence number six containing all three distinguished vertices. An explicit fractional coloring of M_4 gives $\rho(M_6) \leq 10/3$; a small exact certificate for a twenty-vertex witness gives equality. The independently verified finite census consists of 1,990 maximizing subsets in 199 orbits under the full ambient group D_5 . We separate the analytic upper and equality reductions from the finite witness and census checks. The strict fractional-coloring hypothesis is a sufficient condition used by the proof; its necessity is not asserted.

1 Statements and scope

For a nonempty finite graph H , its Hall ratio is

$$\rho(H) = \max_{\emptyset \neq S \subseteq V(H)} \frac{|S|}{\alpha(H[S])}.$$

We abbreviate $\alpha(H[S])$ to $\alpha(S)$ when the ambient graph is clear. The ordinary Mycielskian $\mu(G)$ contains original vertices $V(G)$, an independent clone layer $V(G)'$, and an apex z . Originals retain their edges, u is adjacent to v' exactly when $uv \in E(G)$, and the apex is adjacent to every clone.

Theorem 1.1. *Let G be a nonempty finite simple graph with $\chi_f(G) < 3$. Then*

$$\rho(\mu(G)) \leq 3, \quad \rho(\mu^2(G)) \leq \frac{10}{3}. \tag{1}$$

Any induced subset attaining $10/3$ in $\mu^2(G)$ has order 20 and independence number 6, and contains all three distinguished apex vertices of the two construction steps. In the four base layers defined below, it satisfies $p + r = 11$ and $q + t = 6$. Its final original and clone cardinalities are one of $(13, 6)$, $(14, 5)$, $(15, 4)$.

We use $M_2 = K_2$ and $M_{r+1} = \mu(M_r)$, giving orders 2, 5, 11, 23, 47. An explicit fractional coloring of M_4 has weight $29/10 < 3$. Thus Theorem 1.1 applies to $M_6 = \mu^2(M_4)$.

Theorem 1.2 (Exact finite specialization). *The graph M_6 has Hall ratio $10/3$. Exactly 1,990 labelled subsets attain this value. They form 199 orbits under $\text{Aut}(M_6) \cong D_5$, each of size ten. The orbit counts by final original/clone/apex sizes are 109 for $(13, 6, 1)$, 83 for $(14, 5, 1)$, and 7 for $(15, 4, 1)$.*

The upper bound and the necessary equality conditions are analytic. The matching lower witness and the complete orbit count use exact finite verification, described in Sections 4 and 5. The orbits are those of embedded subsets under the ambient group; they are not asserted to be distinct abstract graph-isomorphism types.

Cropper, Gyárfás and Lehel [1] establish the known Hall values through M_5 and unboundedness of the Mycielski sequence. Fractional coloring under this construction is classical [2]; the Hall-profile framework also predates this note [3]. Steiner [4] records the separate open question of boundedness of $\chi_f(M_r)/\rho(M_r)$. The present result does not resolve that asymptotic question. We make no first-occurrence claim for the general inequality or the finite census. The hypothesis $\chi_f(G) < 3$ is sufficient for this proof; no necessity or boundary-threshold claim is made.

2 Fractional coloring and the four-layer model

A fractional coloring is a collection of nonnegative weights on independent sets such that every vertex has coverage at least one. Its total weight bounds the fractional chromatic number. Choose a fractional coloring of G of total weight $c < 3$.

Lemma 2.1. *If a graph X admits a homomorphism to G , then*

$$|V(X)| \leq c\alpha(X). \quad (2)$$

Proof. Pull back the weighted independent sets along the homomorphism. They are independent in X and cover every vertex with weight at least one. Summing the coverage gives $|V(X)| \leq \sum_I w_I |I| \leq c\alpha(X)$. This also holds when X is empty. $\square$

Write the vertices of $\mu^2(G)$ as four copies of the base and three special vertices:

$$P, Q, z, R, T, z', w.$$

Here P is the original G , Q its first clone layer, and z the first apex. The second-stage clones of P, Q, z are R, T, z' , respectively; w is the final apex. We use the same letters for the selected vertices in these layers, with sizes p, q, r, t .

Every edge among P, Q, R, T projects to an edge of G by forgetting the layer. The special adjacencies are: z is adjacent to all of Q, T ; z' is adjacent to all of Q and to w ; and w is adjacent to all of R, T, z' . In particular,

$$Q \cup T \text{ is independent,} \quad \{z, z'\} \text{ is independent and has no neighbors in } P \cup R. \quad (3)$$

These elementary facts supply the refined bound below.

Since $\chi_f(G) < 3$, the graph G is triangle-free: a triangle would require coloring weight at least three. The Mycielski construction preserves triangle-freeness. A triangle containing the apex is impossible because its neighborhood is independent; a triangle containing a clone would project to a triangle in the base.

3 Proof of the general bounds and equality restrictions

3.1 One construction step

Let U be a nonempty induced subset of $\mu(G)$ with independence number a . Deleting its apex leaves a graph homomorphic to G , so

$$|U| \leq ca + 1 < 3a + 1.$$

Integrality gives $|U| \leq 3a$, proving $\rho(\mu(G)) \leq 3$.

3.2 Two construction steps

Let S be a nonempty induced subset of $\mu^2(G)$, set $k = \alpha(S)$, and put $Z = \{z, z', w\}$. Deleting these three vertices and applying Lemma 2.1 gives

$$|S| \leq \lfloor ck \rfloor + 3 \leq 3k + 2. \quad (4)$$

For every integer $k \geq 6$, $3k + 2 \leq 10k/3$, strictly when $k \geq 7$.

For $3 \leq k \leq 5$, distinguish two cases. If at most two vertices of Z are present, then

$$|S| \leq \lfloor ck \rfloor + 2 \leq 3k + 1. \quad (5)$$

If all three are present, set $X = P \cup R$ and $Y = Q \cup T$. By (3), $|Y| \leq k$ and $\alpha(X) \leq k - 2$. Consequently

$$|S| \leq \lfloor c(k - 2) \rfloor + k + 3 \leq 4k - 4. \quad (6)$$

The resulting bounds for $k = 3, 4, 5$ are 10, 13, 16, respectively, all at most $10k/3$.

The remaining cases use only triangle-freeness. Independence number one permits at most two vertices. Every triangle-free graph on six vertices has an independent triple: three neighbors of a vertex form one; otherwise that vertex has three nonneighbors, among which a nonedge together with the vertex gives one. Thus independence number two permits at most five vertices. This proves both inequalities in (1).

3.3 Equality conditions

All cases are strict in ratio except possibly $k = 3$ or $k = 6$. A triangle-free graph on ten vertices contains an independent four-set: a vertex of degree at least four supplies one in its neighborhood; otherwise any vertex has at least six nonneighbors, whose independent triple can be enlarged by that vertex. Thus $k = 3$ cannot attain $10/3$. Equality requires $k = 6$ and $|S| = 20$.

If at most two distinguished vertices were present, the same deletion argument would give $|S| \leq \lfloor 6c \rfloor + 2 \leq 19$. Hence all of z, z', w are selected. Equation (3) gives

$$|X| \leq \lfloor 4c \rfloor \leq 11, \quad |Y| \leq 6.$$

Since $|X| + |Y| = 17$, both bounds are tight:

$$p + r = 11, \quad q + t = 6. \quad (7)$$

Moreover $\alpha(X) = 4$, since $|X| = 11$ and $c < 3$, while Y is an independent six-set.

Let A be the final original part of S , an induced subset of $\mu(G)$, and B its final clone-index set. The selected apex w is nonadjacent to A , so $\alpha(A) \leq 5$. The one-step result gives $|A| \leq 15$. Since B' is independent, $|B| \leq 6$, and $|A| + |B| = 19$ implies $|A| \geq 13$. If $\alpha(A) \leq 4$, then the same one-step result would give $|A| \leq 12$, a contradiction. Therefore $\alpha(A) = 5$ and the three final layer sizes in Theorem 1.1 follow.

4 The sixth Mycielski graph

4.1 An explicit coloring below three

There is no need to import an exact fractional chromatic value for this application. Let C_5 have its five independent pairs I . In $M_4 = \mu(C_5)$, assign the following independent-set weights:

Independent sets	Number	Weight per set
$\{z\} \cup I$	5	1/5
$I \cup I'$	5	3/10
The full clone layer	1	2/5

Every cycle vertex belongs to two independent pairs. Thus an original vertex has coverage $2(1/5 + 3/10) = 1$, a clone has coverage $2(3/10) + 2/5 = 1$, and the apex has coverage $5(1/5) = 1$. The total weight is $29/10 < 3$. Theorem 1.1 therefore proves

$$\rho(M_6) \leq 10/3. \quad (8)$$

This elementary construction is consistent with the classical fractional coloring formula of Larsen, Propp and Ullman [2].

4.2 A local exact certificate for attainment

At each construction step, label originals $0, \dots, n-1$, clones $n, \dots, 2n-1$, and the apex $2n$. In M_6 take

$$W = \{0, 1, 2, 3, 4, 5, 6, 7, 11, 12, 15, 19, 21, 22, \\ 26, 32, 33, 41, 45, 46\}.$$

The set $\{11, 12, 15, 19, 21, 41\} = W \cap (Q \cup T)$ is independent of size six. An exact finite certificate proves $\alpha(M_6[W]) \leq 6$.

For completeness, the certificate uses only the recurrence

$$a(\emptyset) = 0, \quad a(U) = \max\{a(U \setminus \{v\}), 1 + a(U \setminus N[v])\}, \quad (9)$$

where v is the least-labelled vertex of nonempty U . The two terms correspond to omitting or including v , so induction proves that $a(U) = \alpha(M_6[U])$. The supplied table contains 190 states, including the empty state and W , and gives $a(W) = 6$. Every nonempty row is checked against its two strictly smaller child states. A separately implemented include/exclude computation also obtains six. No floating-point calculation or full 47-vertex optimization is needed for this witness.

Thus $|W|/\alpha(W) = 20/6 = 10/3$, proving equality in (8). The constant in the double-step bound is therefore attained within the stated class, at $G = M_4$. This attainment does not assert necessity of the strict hypothesis $\chi_f(G) < 3$.

5 The complete finite census

The analytic equality restrictions reduce the census to final original sets $A \subseteq V(M_5)$ of orders 13, 14, 15 with $\alpha(A) = 5$, clone-index sets B of size $19 - |A|$, and the final apex. Here $N(I)$ denotes the open neighborhood of I in M_5 . For such a pair,

$$\alpha(M_6[A \cup B' \cup \{w\}]) = \max \left\{ 1 + \alpha(A), \max_{\substack{I \subseteq A \\ I \text{ independent}}} (|I| + |B \setminus N(I)|) \right\}. \quad (10)$$

Indeed, an independent set using w contains no clone; one avoiding w chooses independent originals I and may take all available clones. Consequently the exact membership criterion is

$$|B \setminus N(I)| \leq 6 - |I| \quad \text{for every independent } I \subseteq A. \quad (11)$$

This criterion does not by itself give a short derivation of the orbit counts below; those counts come from complete finite enumeration.

5.1 The ambient group

The initial M_3 is C_5 , with cyclic order $(0, 1, 2, 4, 3)$ in the recursive labels. At every later stage the apex is the unique maximum-degree vertex. If the base order is n and its maximum degree is $(n - 1)/2$, the new original degrees are at most $n - 1$, clone degrees at most $(n + 1)/2$, and apex degree n ; this property inducts from C_5 .

An automorphism fixes the apex and preserves the original and clone layers. Its original restriction is a base automorphism. The bases are open-neighborhood twin-free, beginning with C_5 , so the clone permutation is uniquely forced by that restriction. Conversely, every base automorphism extends. Thus $\text{Aut}(M_6) = D_5$, of order ten. The independent check obtains the ten cycle automorphisms by testing all 120 permutations and then verifies their lifted adjacency preservation.

5.2 Two independently checked complete enumerations

The primary program computes $\alpha(A)$ on the 2^{23} base subsets using (9). It retains only the specified original sets and one representative under D_5 , leaving 4,370 candidates. For each A , choose an independent five-set $I_0 \subseteq A$ minimizing $|N(I_0)|$. Equation (11) requires at most one clone index outside $N(I_0)$. Thus enumerate exactly the $b = 19 - |A|$ element sets B consisting of either b elements of $N(I_0)$, or $b - 1$ elements of $N(I_0)$ and one outside. Test all remaining inequalities and canonicalize the full subset.

This examines 50,504,405 filtered clone candidates. It accepts 200 pairs at canonical original sets, which reduce to 199 full-subset orbits. The last deduplication removes the action of an original-set stabilizer on clone choices.

The independent implementation uses a maximum subset-zeta transform for the base independence values, and recursive capacity propagation of (11) for clone choices. Its 184,588 search nodes produce exactly the same 199 canonical masks. Each pruning rule is checked for logical sufficiency. Every accepted subset is retested against the complete independently generated list of 857 maximal independent sets of M_6 .

All 1,990 labelled output sets also satisfy the new analytic restrictions: they contain all three distinguished vertices and have the split (7). Their stabilizers are trivial. The resulting census is

Originals	Clones	Apex	Orbits	Labelled subsets
13	6	1	109	1,090
14	5	1	83	830
15	4	1	7	70
Total			199	1,990

The analytic reductions prove that no other sizes or layers are omitted; the exact searches establish the count inside the permitted classes.

6 Verification boundary and prior work

The proof of Theorem 1.1, the explicit 29/10 coloring, and the necessary equality reductions are analytic. The lower witness has the small recurrence certificate (9); the census remains computer-assisted. The earlier 273-node upper-bound certificate and 22-node equality-exclusion certificate are preserved as verification history, but are no longer premises of the proof.

The source package includes the analytic note, the explicit fractional coloring, the 190-state witness table and its standard-library checker, and an independent audit. The earlier census programs, complete orbit data and independent capacity-propagation check are preserved separately. No generated format files, caches or page images are needed for the mathematical reproduction.

The computation uses standard finite recurrences and enumeration. No improved general maximal-independent-set algorithm is claimed: Tsukiyama et al. [5] already give output-sensitive enumeration in $O(NMK)$ time for a graph with N vertices, M edges and K maximal independent sets. That row-generation problem is distinct from the exact Hall optimization and extremizer census.

A bounded current-source check examined the specific $\chi_f(G) < 3$ one-step and two-step Hall statements, including Barnett’s Chapter 4 [3]. It did not locate an equivalent statement in the inspected material. This is not an exhaustive priority review. The present wording is “we prove”, not a first-ever claim. Neither the general bound nor the finite result resolves the asymptotic question discussed in [4].

中文摘要

对任意非空有限简单图 G , 若 $\chi_f(G) < 3$, 则本文证明 $\rho(\mu(G)) \leq 3$ 及 $\rho(\mu^2(G)) \leq 10/3$. 上界证明只使用分数着色、三个特殊顶点的删除以及初等独立集论证, 不依赖有限分支证书。第二个不等式取等号时, 诱导顶点集必有 20 个顶点、独立数 6, 并包含全部三个特殊顶点。显式的 29/10 分数着色使该定理适用于 M_6 ; 一个 20 顶点见证的 190 状态精确递归证书给出匹配下界。全部 1990 个极值顶点集在环境图的自同构群 D_5 下分成 199 个轨道, 轨道计数仍由两个独立的有限枚举程序验证。严格假设 $\chi_f(G) < 3$ 是本证明采用的充分条件, 未断言其必要性或阈值最优性; 本文也不作首次发表或渐近问题已解决的声明。

References

- [1] M. Cropper, A. Gyárfás and J. Lehel, *Hall ratio of the Mycielski graphs*, Discrete Mathematics **306** (2006), 1988–1990. doi:10.1016/j.disc.2005.09.020.
- [2] M. Larsen, J. Propp and D. Ullman, *The fractional chromatic number of Mycielski’s graphs*, Journal of Graph Theory **19**(3) (1995), 411–416. doi:10.1002/jgt.3190190313.
- [3] J. B. Barnett, *The Fractional Chromatic Number and the Hall Ratio*, Ph.D. dissertation, Auburn University, 2016. Auburn dissertation repository, Definition 2.1.1 and Chapter 4.
- [4] R. Steiner, *Fractional Chromatic Number Vs. Hall Ratio*, Combinatorica **45** (2025), article 37. doi:10.1007/s00493-025-00164-0.
- [5] S. Tsukiyama, M. Ide, H. Ariyoshi and I. Shirakawa, *A New Algorithm for Generating All the Maximal Independent Sets*, SIAM Journal on Computing **6**(3) (1977), 505–517. doi:10.1137/0206036.